Project Report

EE 517 Fall 2024

HW 4 — RCS

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In this project, I modeled 10 complex targets as a set of 50 point-scatterers that are randomly distributed in a box that is ±5 meters in the x dimension (along the initial radar boresight) and ±2.5 meters in the y dimension, centered at (x,y) = (0,0). Scatterers are located at (x,y) coordinate pairs.

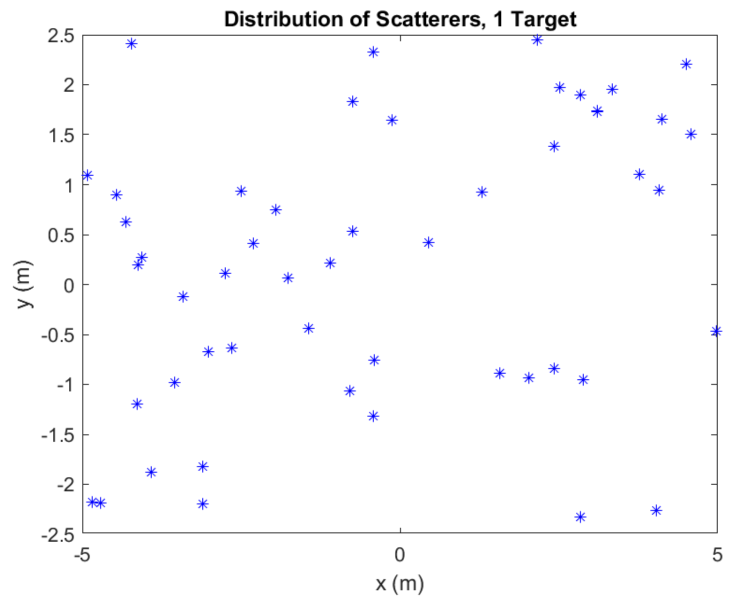

Figure 1. The blue asterisks show the distribution of 50 scatterers of fixed RCS = 1 m^2 each for 1 target.

I simulated a monostatic radar, with a transmit frequency of 10 GHz, on an airborne vehicle orbiting a target located 10km away from the center of the target. The total RCS of the target is calculated at each aspect angle from 0° to 359° in 0.5° intervals, by summing each relative RCS of each scatters at each angle. The db RCS value was calculated and plotted against the aspect angle. The mean of

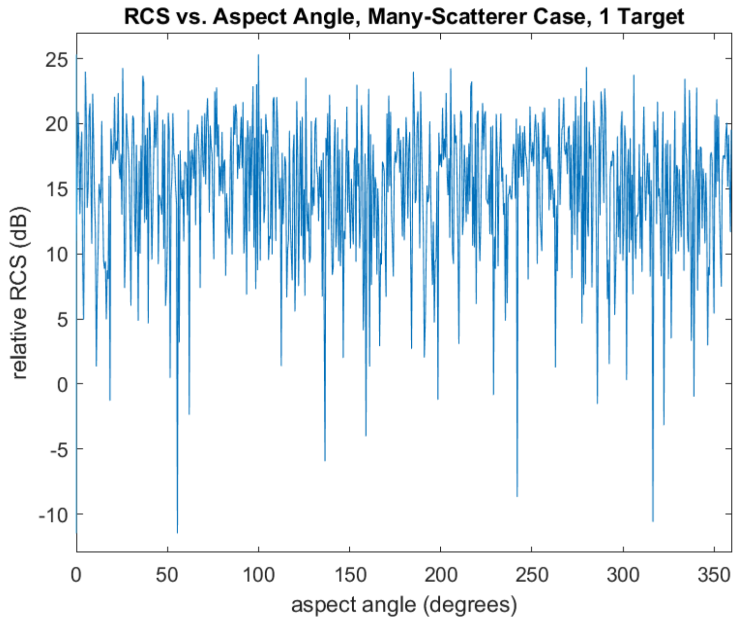

Figure 2. Relative RCS of a complex target vs. aspect angle at a range of 10 km and transmit frequency of 10 GHz.

The RCS fluctuations shown in Figure 2 make the use of a statistical description of the RCS more convenient. A specific exponential probability density function (PDF) can model the RCS of the scatters within a single-resolution cell. The PDF used for this project was for the use case of a target that has many scatterers, randomly distributed, and none being dominant, which matched the parameters of the modeled targets. I calculated and plotted this PDF for all complex targets. For the PDF, I had to calculate the mean of the linear scale RCS. In this case, the mean of the RCS values would approximate the individual scatter's RCS as the number of scatters increases. Additionally, I modeled a histogram of RCS data with varying aspect angles for the targets, which matched the PDF. The RCS data for this histogram was normalized.

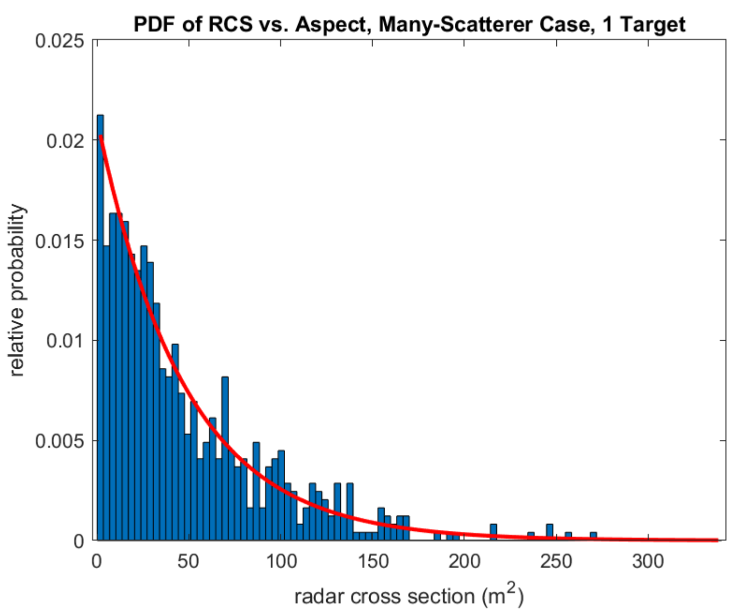

Figure 3. Histogram of RCS data with varying aspect angles and exponential PDF with the same RCS mean for a single target

I computed the average RCS data histogram over 10 different targets. This gave a better approximation of the theoretical exponential PDF than the single target histogram because as the number of samples increases the values tend toward the theoretical values.

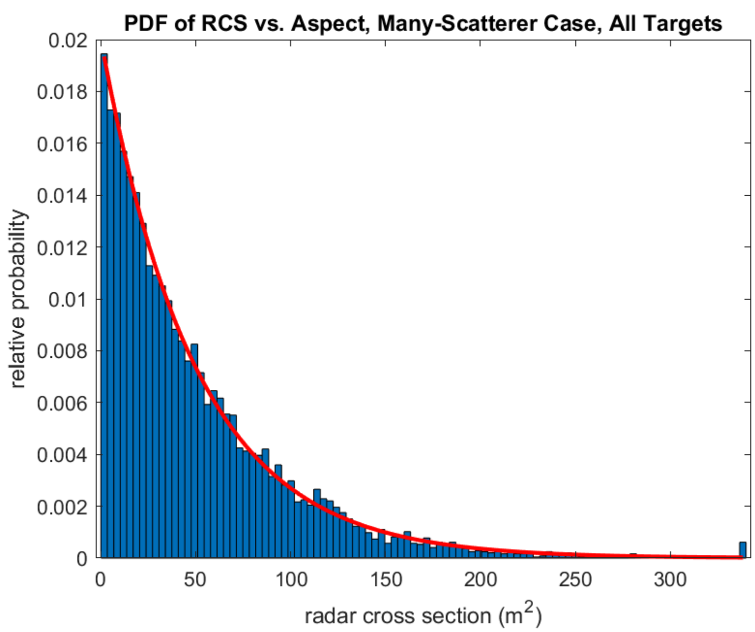

Figure 4. Averaged histogram of the RCS data with varying aspect angles and exponential PDF with the same RCS mean.

For 10 targets, I calculated the RCS for 10 targets at a fixed aspect angle, but varying RF frequency. Just like the aspect angle, a varying RF frequency causes RCS fluctuations. This was modeled in a histogram of normalized RCS values with a varying frequency from 10 GHz to 14.5 GHz in 15 MHz intervals. Next, the RCS data histograms of 10 targets were averaged into one histogram. Along with the theoretical exponential PDF for our specific complex targets. The averaged 10-target histogram better matched the PDF, than the single-target histogram. This is due to the averaged histogram having more sample values which causes the histogram to tend more towards the PDF.

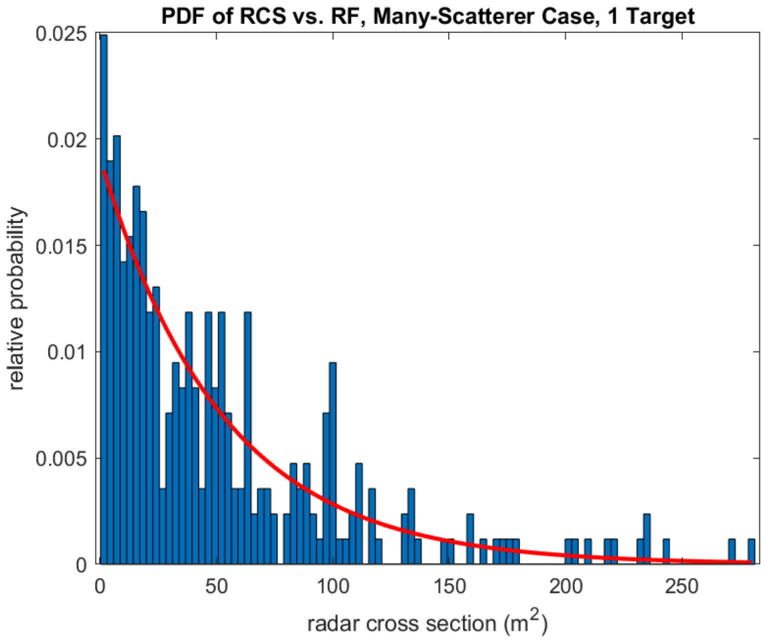

Figure 5. Histogram of RCS data with varying RF frequency and exponential PDF with the same RCS mean for a single target

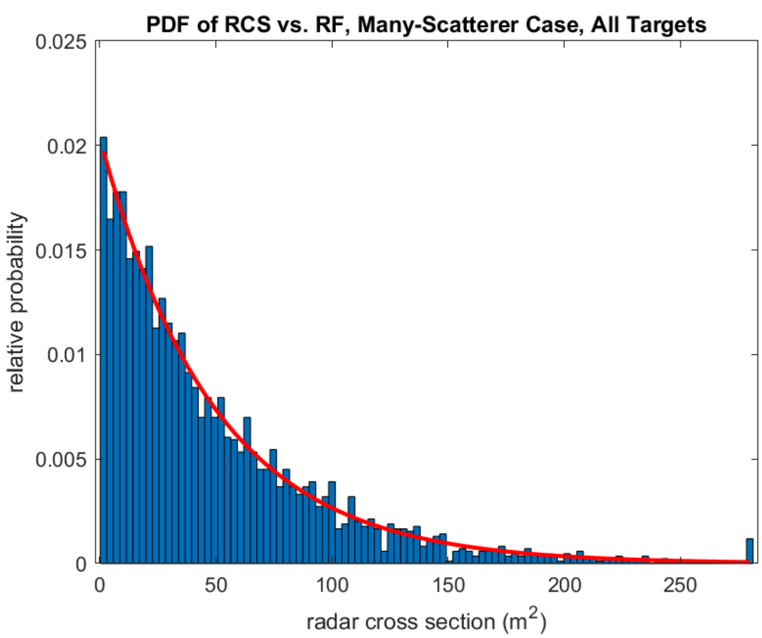

Figure 6. Averaged Histogram of RCS data with varying RF frequency and exponential PDF with the same RCS mean.